

Goals

Flooder: as many messages sent as possible; the service pays per message.

Pumper: as many messages as possible to its partner's ranges; it is paid a share of the termination fee and earns nothing elsewhere.

Cannot do

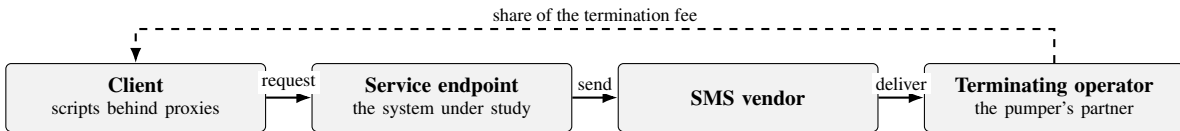
forge an app attestation; earn on a range its partner does not terminate; make a stranger enter a code, so a code entered on a block real users share is honest evidence.

Outside the model

attacks on a code after delivery (Section 2.5); compromise of the service, its vendors or the attestation providers.

Instantiated by

the 25 attacker profiles of Table 2; pool size, rate and CAPTCHA class are drawn per seed



Controls at the client

residential addresses in-country, a fresh one per request, or a datacenter abroad, or 20 ASNs at a low rate; human-like reCAPTCHA v3 tokens bought at about 0.3 cents; fingerprints minted fresh, aged for hours, or one profile reused; a forged platform header; numbers drawn at random, walked upward, or taken from its partner's blocks

Knows about the service

every threshold, the test parameters and the workers' timing; observes one uniform response body, after at least 400 ms (Section 3.3)

Cannot touch

the vendor's route, receipts for numbers it does not control, or the service's state

Controls on its own ranges

enters codes within a second or after a human-like delay, or only when a block nears its threshold; reports deliveries as failed; verifies on 500 identity-and-number pairs for 10 or 30 minutes before flooding; spreads over 3 to 300 blocks, at full rate or paced to a quota; buys a solution for every interactive challenge